

Tools of the Trade: Linux and SQL

Module 1
Introduction to operating systems

Get started with the course

Introduction to Course 4

Video • 2 min

Course 4 overview

Reading • 10 min

Helpful resources and tips

Reading • 8 min

The wonderful world of operating systems

Welcome to module 1

Video • 52 sec

Kim: My journey into computing

Video • 1 min

Introduction to operating systems

Video • 2 min

Compare operating systems

Reading • 8 min

Test your knowledge: The wonderful world of operating systems

Practice Assignment • Grade: 100%

The operating system at work

Inside the operating system

Video • 3 min

Requests to the operating system

Reading • 8 min

Practice: Order of operations in the OS

Ungraded Plugin • 10 min

Resource allocation via the OS

Video • 2 min

Virtualization technology

Reading • 8 min

Test your knowledge: The operating system at work

Practice Assignment • Grade: 100%

The user interface

GUI versus CLI

Video • 4 min

The command line in use

Reading • 4 min

Activity: Select the user interface

Practice Assignment • Grade: 100%

Ellen: My path into cybersecurity

Video • 2 min

Test your knowledge: The user interface

Practice Assignment • Grade: 100%

Review: Introduction to operating systems

Wrap-up

Video • 1 min

Glossary terms from module 1

Reading • 2 min

Module 1 challenge

Graded Assignment • Grade: 100%

Module 2
The Linux operating system

All about Linux

Welcome to module 2

Video • 56 sec

Introduction to Linux

Video • 3 min

Phil: Learn and grow in the cybersecurity field

Video • 2 min

Linux architecture

Video • 3 min

Linux architecture explained

Reading • 8 min

Explore: Components of the Linux OS

Ungraded Plugin • 10 min

Test your knowledge: All about Linux

Practice Assignment • 8 min

Linux distributions

Linux distributions

Video • 2 min

KALI LINUX™

Video • 2 min

More Linux distributions

Reading • 4 min

Package managers for installing applications

Reading • 8 min

Resources for completing Linux labs

Reading • 4 min

Lab tips and troubleshooting steps

Reading • 4 min

Activity: Install software in a Linux distribution

Ungraded App Item • 30 min

Optional Exemplar: Install software in a Linux distribution

Ungraded App Item • 10 min

Exemplar: Install software in a Linux distribution

Reading • 8 min

Test your knowledge: Linux distributions

Practice Assignment • 8 min

The shell

Introduction to the shell

Video • 1 min

Different types of shells

Reading • 4 min

Input and output in the shell

Video • 2 min

Activity: Examine input and output in the shell

Ungraded App Item • 30 min

Optional Exemplar: Examine input and output in the shell

Ungraded App Item • 10 min

Exemplar: Examine input and output in the shell

Reading • 10 min

Test your knowledge: The shell

Practice Assignment • 8 min

Review: The Linux operating system

Wrap-up

Video • 51 sec

Linux architecture explained

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Understanding the Linux architecture is important for a security analyst. When you understand how a system is organized, it makes it easier to understand how it functions. In this reading, you'll learn more about the individual components in the Linux architecture. A request to complete a task starts with the user and then flows through applications, the shell, the Filesystem Hierarchy Standard, the kernel, and the hardware.

User

The **user** is the person interacting with a computer. They initiate and manage computer tasks. Linux is a multi-user system, which means that multiple users can use the same resources at the same time.

Applications

An **application** is a program that performs a specific task. There are many different applications on your computer. Some applications typically come pre-installed on your computer, such as calculators or calendars. Other applications might have to be installed, such as some web browsers or email clients. In Linux, you'll often use a package manager to install applications. A **package manager** is a tool that helps users install, manage, and remove packages or applications. A **package** is a piece of software that can be combined with other packages to form an application.

Shell

The **shell** is the command-line interpreter. Everything entered into the shell is text based. The shell allows users to give commands to the kernel and receive responses from it. You can think of the shell as a translator between you and your computer. The shell translates the commands you enter so that the computer can perform the tasks you want.

Filesystem Hierarchy Standard (FHS)

The **Filesystem Hierarchy Standard (FHS)** is the component of the Linux OS that organizes data. It specifies the location where data is stored in the operating system.

A **directory** is a file that organizes where other files are stored. Directories are sometimes called "folders," and they can contain files or other directories. The FHS defines how directories, directory contents, and other storage is organized so the operating system knows where to find specific data.

Kernel

The **kernel** is the component of the Linux OS that manages processes and memory. It communicates with the applications to route commands. The Linux kernel is unique to the Linux OS and is critical for allocating resources in the system. The kernel controls all major functions of the hardware, which can help get tasks expedited more efficiently.

Hardware

The **hardware** is the physical components of a computer. You might be familiar with some hardware components, such as hard drives or CPUs. Hardware is categorized as either peripheral or internal.

Peripheral devices

Peripheral devices are hardware components that are attached and controlled by the computer system. They are not core components needed to run the computer system. Peripheral devices can be added or removed freely. Examples of peripheral devices include monitors, printers, the keyboard, and the mouse.

Internal hardware

Internal hardware are the components required to run the computer. Internal hardware includes a main circuit board and all components attached to it. This main circuit board is also called the motherboard. Internal hardware includes the following:

- The **Central Processing Unit (CPU)** is a computer's main processor, which is used to perform general computing tasks on a computer. The CPU executes the instructions provided by programs, which enables these programs to run.
- Random Access Memory (RAM)** is a hardware component used for short-term memory. It's where data is stored temporarily as you perform tasks on your computer. For example, if you're writing a report on your computer, the data needed for this is stored in RAM. After you've finished writing the report and closed down that program, this data is deleted from RAM. Information in RAM cannot be accessed once the computer has been turned off. The CPU takes the data from RAM to run programs.
- The **hard drive** is a hardware component used for long-term memory. It's where programs and files are stored for the computer to access later. Information on the hard drive can be accessed even after a computer has been turned off and on again. A computer can have multiple hard drives.

Key takeaways

It's important for security analysts to understand the Linux architecture and how these components are organized. The components of the Linux architecture are the user, applications, shell, Filesystem Hierarchy Standard, kernel, and hardware. Each of these components is important in how Linux functions.

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Understanding the Linux architecture is important for a security analyst because it helps in comprehending how the system functions.

Core Components of Linux Architecture

- User:** The individual interacting with the computer, initiating and managing tasks in a multi-user system.
- Applications:** Programs that perform specific tasks, installed either pre-installed or via a package manager.

Intermediaries and Organization

- Shell:** The command-line interpreter that translates user commands for the kernel and receives responses.
- Filesystem Hierarchy Standard (FHS):** Organizes data by specifying where files and directories are stored within the operating system.

System Management and Hardware

- Kernel:** Manages processes and memory, communicating with applications and controlling hardware functions.
- Hardware:** The physical components of a computer, categorized as peripheral (e.g., monitors, keyboards) or internal (e.g., CPU, RAM, hard drive).

And if you want to continue exploring this topic, try one of these follow-up questions:

What is the role of the Linux kernel in system management?

How can you use the shell to interact with the Linux kernel?

How does the Filesystem Hierarchy Standard improve system organization and security?

Go to next item →

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